Refer to the content - https://drive.google.com/drive/folders/16AhbcDX0v1n0UT9oS1n_hejnv1ldPVBa?usp=drive_link

Chapter 01: Introduction Chapter

Research gap
Contribution to BoK
1. Contribution to Problem Domain
2. Contribution to Technical Domain

Research Challenge
Research Questions
Research Aim – one sentence – Design, Develop and evaluate
Research Objectives – list of objectives to achieve the main – like a checklist

Chapter 02: Literature Review

2.1 Chapter Overview
2.2 Concept Graph
2.3 Problem Domain
2.4 Technological Review for IMPL project / for CSF students who do NON IMPL project Methodological Review
2.5 Existing Work
2.6 Evaluation and Benchmarking (may not be applicable for CSF students who do NON IMPL project)
2.7 Chapter Summary

How to start LR

Find minimum 20 research paper
On average 40 research paper
Good is 60 research paper
Ideal is all the paper published within last 5 years

Whatever the number of research paper you are going to refer to has be recent research paper

Just to pass => 50 marks => 10 research paper will

Now you need to read the whole research paper => no more just reading (abstract -> introduction -> conclusion (Future work) + Methodology) now read and summarise the whole research paper

Prompt (you can make it better)

Now I'm writing the literature review and my project coordinator recommended me to read the whole research paper and summarise them in a indepth and detailed manner with critical review in tabular form which is he said literature survey he also said Now you need to read the whole research paper => no more just reading (abstract -> introduction -> conclusion (Future work) + Methodology) now read and summarise the whole research paper   my research topic is "Heart attack prediction system using AI" so can you give me a detailed tabular structure to summarise every important aspect of a research paper. Give me tabular format to summarise which I can copy and paste to spread sheet https://docs.google.com/spreadsheets/d/1NDJ-mUV1yDDhNh4TIMRmiP111610cIlPnYkh2hwiu34/edit?usp=sharing

https://chatgpt.com/share/69a2efe4-8b08-8011-990b-9968721c088d

Read a research paper cannot understand? Use notebook LM / any AI tool to summarize / visualize understand never rely on AI tool once you get the understanding summarize on your own put it into a tabular form

Use this tabular form - https://docs.google.com/spreadsheets/d/1NDJ-mUV1yDDhNh4TIMRmiP111610cIlPnYkh2hwiu34/edit?usp=sharing

How to come up with the tabular form - https://chatgpt.com/share/69a2efe4-8b08-8011-990b-9968721c088d

These tabular forms are not fixed (Not Bible!) therefore you can improve iteratively and you will be reading research paper back and forth

The summary is Literature Survey not Literature Review

Literature Review = Literature Survey + **critiquing**

When writing LR we will use thematic analysis – use an AI to get a better understanding

Good — you already have **excellent guidelines**.
What you need now is **how to adapt that structure specifically for**:

Heart Attack Prediction using Artificial Intelligence

Below is a **research-ready Chapter 2 structure**, mapped **exactly to your guideline sections**, but rewritten so every subsection directly supports your topic and leads logically into your methodology.

This is **NOT generic** — it is tailored for **medical AI prediction systems**.

✅ CHAPTER 2 – LITERATURE REVIEW

Topic: *Heart Attack Prediction using AI*

Target Length: ~15 pages
Goal: Build a logical argument:

Cardiovascular prediction problem → AI solutions → limitations → research gap → your system.

2.1 Chapter Overview (≈ ½ page)

Purpose

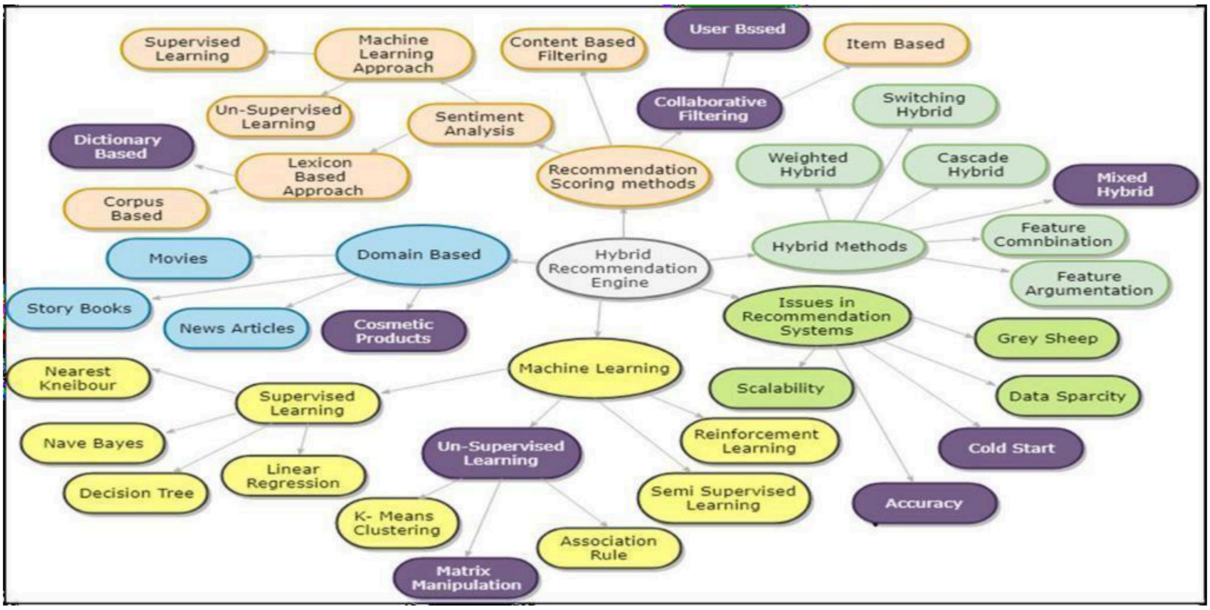
Explain:

- Why heart attack prediction is an important research problem
- Why AI is being used
- What this chapter analyses
- How sections are organised

Example logical flow (adapted)

This chapter presents a critical review of artificial intelligence approaches for heart attack prediction. It begins by mapping the conceptual structure of cardiovascular risk prediction and related AI technologies. The problem domain is then analysed, followed by a technological review of machine learning pipelines used in medical prediction systems. Existing research is examined through thematic analysis, and current evaluation and benchmarking practices are critically assessed. The chapter concludes by identifying research gaps that motivate the proposed methodology.

2.2 Concept Map



~~What your concept map MUST include~~

~~Central Node~~

~~Heart Attack Prediction using AI~~

~~Level-1 branches~~

1. ~~Cardiovascular Disease Knowledge~~
2. ~~Medical Data Sources~~
3. ~~Machine Learning Techniques~~
4. ~~Model Evaluation~~
5. ~~Clinical Deployment Challenges~~
6. ~~Explainable AI~~

~~Level-2 examples~~

~~Medical Data~~

- ~~Clinical records~~
- ~~ECG signals~~
- ~~Lifestyle attributes~~
- ~~Biomarkers~~

~~AI Models~~

- ~~Logistic Regression~~
- ~~Random Forest~~
- ~~Deep Neural Networks~~
- ~~Ensemble Learning~~

~~**Challenges**~~

- ~~Data imbalance~~
- ~~Interpretability~~
- ~~Generalisation~~
- ~~Dataset bias~~

📌 Insert as:

Figure 2.1 Concept Map of AI-Based Heart Attack Prediction

(Full version → Appendix)

2.3 ~~Problem Domain~~ Cardiovascular Risk Prediction (~4 pages)

This section answers:

👉 *What is the medical + research problem?*

2.3.1 Introduction to Cardiovascular Risk Prediction

Discuss:

- Cardiovascular diseases (CVD) overview
- Heart attacks (myocardial infarction)
- Mortality statistics
- Importance of early prediction

Define key concepts:

- Risk prediction
- Clinical decision support systems
- Preventive healthcare analytics

Explain traditional approaches:

- Framingham Risk Score
- Clinical rule-based assessment

👉 Then introduce AI as evolution.

2.3.2 Key Challenges in Heart Attack Prediction

Organise into **4–5 sub-areas**.

(A) Data Heterogeneity

- Different hospitals
- Missing clinical values
- Structured vs physiological data

(B) Class Imbalance

- Few positive heart attack cases
- Bias toward healthy predictions

(C) Model Interpretability

- Doctors require explanations
- Black-box AI limitations

(D) Generalisation & Dataset Bias

- Models trained on small datasets
- Poor cross-population performance

(E) Clinical Reliability

- False negatives are dangerous

Each subsection:

- Define
- Explain importance
- Cite studies
- End with linking sentence

2.3.3 Existing Frameworks / Standards

Discuss:

- Clinical risk scoring systems
- Medical AI ethics
- Data privacy regulations

Examples:

- GDPR (health data)
- HIPAA concepts (if cited internationally)
- WHO digital health guidelines

Explain how these influence AI system design.

2.3.4 Proposed Conceptual Architecture (Positioning) ~ 1 page max

Introduce **your system conceptually** (high level only).

Example pipeline:

Patient Data → Preprocessing → Feature Engineering → AI Model → Explainability → Risk Prediction Dashboard

Insert:

Figure 2.2 Proposed Conceptual Architecture

Explain briefly how architecture addresses earlier challenges.

2.4 Technological Review (~4 pages)

This is NOT algorithm listing.

You review **ML pipeline stages**.

2.4.1 Data Acquisition & Pre-processing

Discuss:

- Medical datasets
- Normalization
- Missing value imputation
- Feature scaling

Critically compare methods.

2.4.2 Feature Engineering Techniques

- Clinical feature importance
- Statistical selection
- PCA/LASSO
- Medical relevance vs statistical relevance

2.4.3 Machine Learning Models for Prediction

Compare categories:

Traditional ML

- Logistic Regression
- SVM
- Random Forest

Ensemble Methods

- XGBoost
- Gradient Boosting

Deep Learning

- ANN models
- Hybrid models

Critically analyse strengths/weaknesses.

2.4.4 Hyperparameter Optimization

- Grid Search
- Random Search
- Bayesian optimization

Explain impact on medical prediction reliability.

2.4.5 Evaluation Metrics

Discuss suitability:

- Accuracy (problematic alone)
- Precision/Recall
- F1-score
- ROC-AUC
- Sensitivity (very important clinically)

2.5 Existing Work (~4 pages)

⚠️ DO NOT summarise paper-by-paper.

Use **themes**.

2.5.1 Theme 1 — Classical Machine Learning Approaches

Compare multiple studies together.

Trends:

- Random Forest dominance
- Tabular datasets

Limitations:

- Limited explainability.

2.5.2 Theme 2 — Deep Learning-Based Prediction

Discuss ANN/DNN systems.

Strengths:

- nonlinear modelling

Weakness:

- data requirements.

2.5.3 Theme 3 — Explainable AI in Healthcare

Discuss SHAP/LIME studies.

Importance:

- clinician trust.

2.5.4 Cross-theme Analysis

Compare:

Aspect	Classical ML	Deep Learning
Data Need	Low	High
Interpretability	High	Low
Performance	Moderate	High

2.6 Evaluation and Benchmarking (~1 pages)

Critically analyse:

Datasets Used

- Cleveland dataset dominance
- Lack of diversity

Benchmark Issues

- Small sample sizes
- Non-clinical validation

Metric Problems

- Overuse of accuracy

Major Evaluation Gap

- Lack of real-world deployment testing.

2.7 Chapter Summary ~ ½ page

Must include:

✅ Main literature themes
✅ Key technical observations
✅ Identified research gaps
✅ Why your research is needed

End with transition:

These identified limitations motivate the methodological framework presented in Chapter 3.

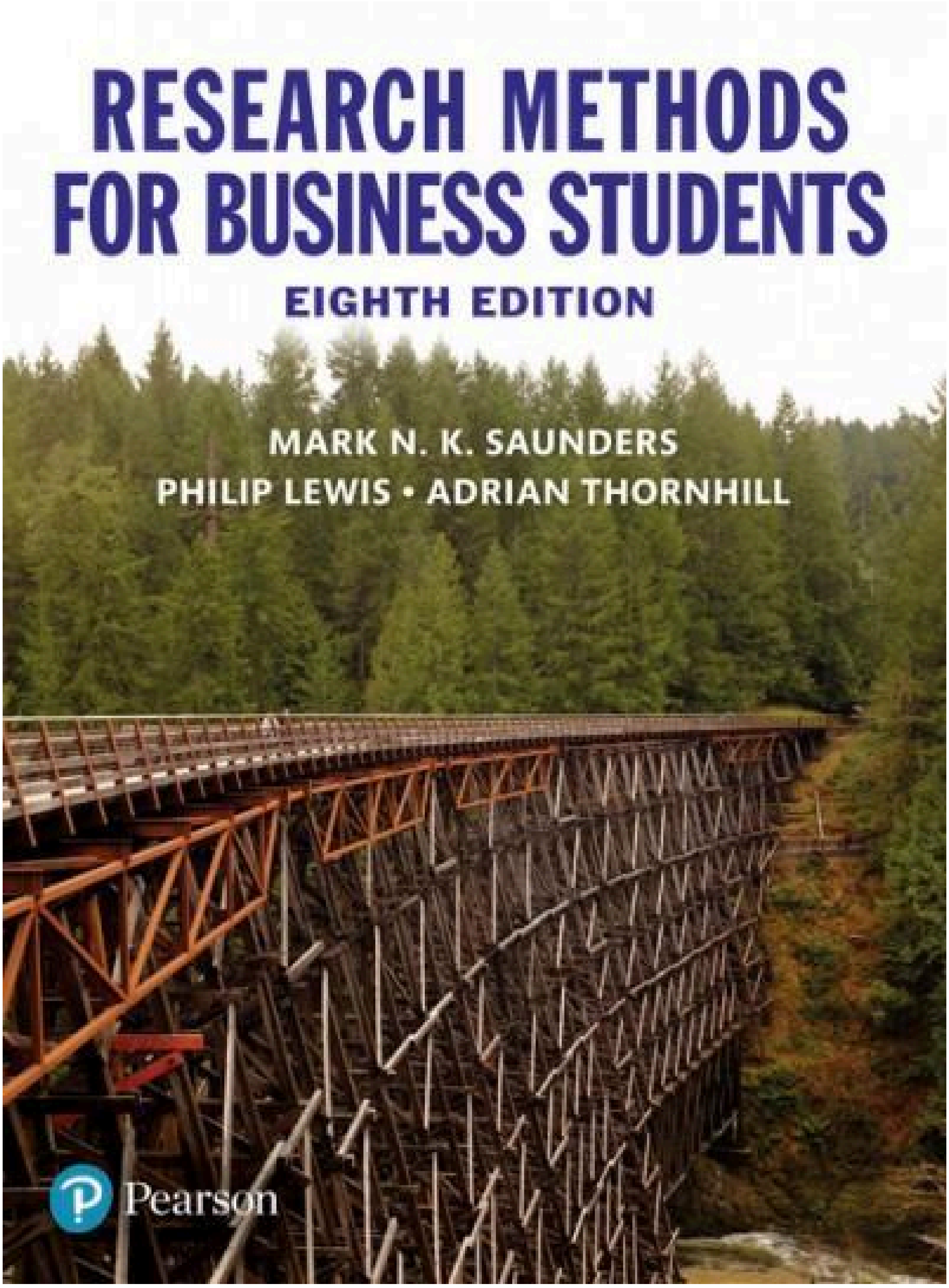
✅ Logical Flow of Your Chapter (Very Important)

```
Medical Problem
        ↓
Prediction Challenges
        ↓
AI Technologies
        ↓
Existing Research
        ↓
Evaluation Weaknesses
        ↓
Research Gap
        ↓
Your Methodology (Chapter 3)
```

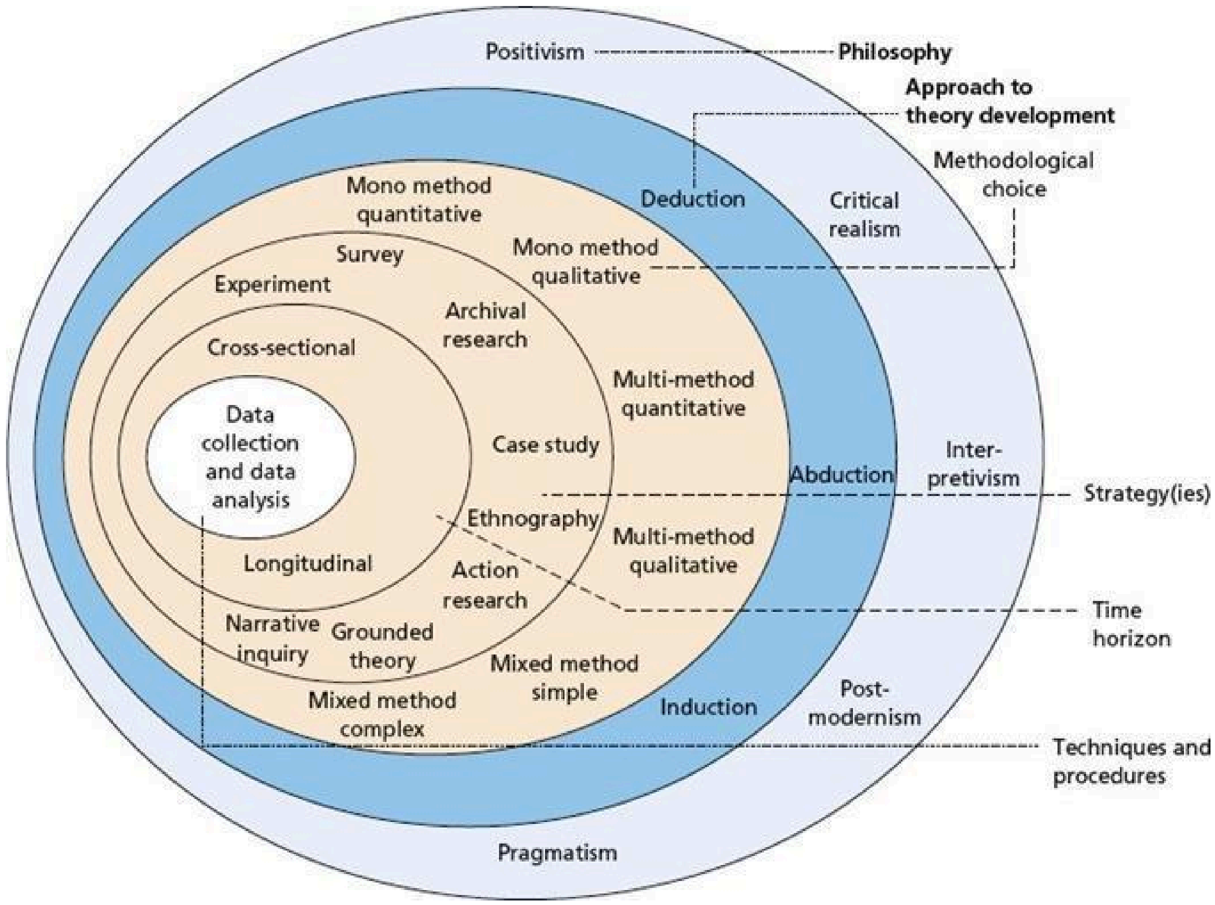
Chapter 03: Methodology

Research methodology

Saunder's research onion



https://gradcoach.com/saunders-research-onion/



Layer	Choice	Justification
Philosophy	Positivism Positivism Interpretivism Pragmatism	
Approach	Deductive Inductive	
Strategy	Experiement Survey Interview Case Study	
Methodological Choice	Mono Method – Quantitative Mono Method – Qual – only interview Multi methods – Quan or Qual	

	Mixed Method	
Time Horizon	Cross sectional Longitudinal	
Data Collection and Analysis		

Development methodology

Development methodology you will use and why (Justification)

Waterfall (not suitable for Software Engineering in general and Research in SE in specific)

Prototyping – trail and error – experiment

Agile method – One Person Scrum

Project management methodology

What is the Project management methodology you will use and why?

Prince 2 or Agile Prince 2

1. Schedule
    - Gantt chart
    - Deliverables and Dates
2. Resources
    - Hardware Resources
    - Software Resources
    - Dataset
    - Skill Requirements
3. Scope
    - In Scope
    - Out Scope
4. Risk and Mitigation

Risk Exposure = Probability of Occurrence * Magnitude of Loss

Probability of Occurrence – Frequency – 1 to 5
Magnitude of Loss – Severity – 1 to 5

Risk Exposure 5 * 5 = 25 – item with Risk Exposure 20 to 25 must be continiously monitored and mitigated

Risk Item	Frequency	Severity	Mitigation Plan